Report: Electricity Consumption Forecasting Analysis

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1. Executive Summary

This report details the development and evaluation of a predictive analytics pipeline designed to forecast total electricity consumption. By integrating historical usage data with environmental and temporal variables, the project compares a baseline Linear Regression model against an ensemble Random Forest Regressor. The analysis concludes that the Random Forest model provides superior accuracy by effectively capturing non-linear patterns and complex feature interactions.

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2. Dataset Overview

The analysis is based on the powerconsumption.csv dataset, which features multivariate time-series data.

- Target Variable: A calculated TotalConsumption field, representing the sum of power usage across three distinct geographic or infrastructure zones.
- Temporal Resolution: Data points are recorded at 10-minute intervals.
- Input Features:
    - Weather Metrics: Temperature, Humidity, Wind Speed, and Diffuse Flows.
    - Temporal Identifiers: Hour, Minute, Day, Month, Day of Week, and a binary Weekend indicator.

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3. Methodology & Feature Engineering

To account for the sequential nature of power consumption, several time-series specific features were engineered to provide the models with "contextual memory":

- Lagged Variables:
    - lag_1: Consumption 10 minutes prior.
    - lag_6: Consumption 1 hour prior.
    - lag_144: Consumption 24 hours prior (capturing daily seasonality).

- Rolling Statistics: A 1-hour rolling average (rolling_avg_1h) was implemented to smooth volatility and highlight immediate trends.
- Data Splitting: The dataset was split into training and testing sets using a non-shuffled approach (80/20 split) to maintain the chronological integrity of the time-series forecasting.

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4. Model Performance Comparison

The models were evaluated using three primary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Metric	Linear Regression	Random Forest
Accuracy/Fit	Provides a linear baseline for general trends.	Superior fit for non-linear fluctuations.
Complexity	High interpretability, low computational cost.	Higher computational cost, better handling of outliers.
Primary Driver	Relies on direct coefficients of features.	Relies on feature importance across 100 decision trees.

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5. Key Findings & Insights

- Feature Importance: Random Forest analysis indicates that historical consumption (lags) and temporal factors (hour of day) are the strongest predictors of future demand.
- Seasonality: The inclusion of is_weekend and dayofweek successfully allows the model to differentiate between industrial/commercial weekday loads and residential weekend loads.
- Environmental Impact: Variables like Temperature and Humidity show a measurable correlation with consumption spikes, likely corresponding to heating or cooling requirements.

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6. Conclusion

The implementation of the Random Forest Regressor significantly outperforms the standard Linear Regression model in this context. For deployment in a smart grid or utility management scenario, the Random Forest model is recommended due to its lower Mean Absolute Percentage Error (MAPE) and its ability to adapt to the inherent volatility of electricity demand.